

Group Comparisons and Statistical Modelling with R

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Chapter overview

Chapter 5 made estimates, intervals and tests executable in base R. Here we return to Chapter 4's group comparisons and use R to connect a fitted model with the question it answers. We begin with three means, compare selected differences, and then examine a six-cell pattern. A separate example adds a pretest so that a group comparison can hold baseline performance fixed.

Counts require a different response model. They also provide a small example of likelihood, which makes a fitting criterion explicit. Finally, simulation examines procedures under a stated generating process, and planning relates sample size to either power or interval width. Throughout, a useful result keeps its data, target, uncertainty calculation and conditions together.

Chapter 5 made estimates, intervals and tests executable for one mean and for one correlation. Studies rarely stop at one mean. Twelve pupils of the invented feedback study practised in the massed schedule, four in each of three feedback conditions, and their three group means on the 20-item assessment are 6, 8 and 10 correct answers. Means that differ in a record differ for more than one reason, so the question is what a stated model and its conditions let us say about three means at once, and what a single answer to that question leaves unsettled. These pupils are authored teaching inputs allocated by stipulation, so each analysis shows what a procedure reports, not what feedback does.

1 Fit and read group means

1.1 Begin with the question and records

Does mean assessment performance differ across three feedback conditions? Chapter 4 developed this question with an invented study of 24 pupils. Each pupil was individually assigned to one combination of feedback (Rereading, Answer feedback or Explanatory feedback) and practice (Massed or Spaced). The outcome is the number correct on a 20-item assessment. Four pupils contribute scores to each cell in Table 1.

Table 1

The same 24 authored scores as Chapter 4.

Feedback	Massed scores	Spaced scores
Rereading	4, 5, 7, 8	4, 5, 7, 8
Answer feedback	6, 7, 9, 10	10, 11, 13, 14
Explanatory feedback	8, 9, 11, 12	16, 17, 19, 20

These values were constructed to make the calculations transparent. They are teaching inputs, not empirical evidence that one feedback condition works better. Each score comes from a different pupil; the four scores within a cell are not repeated measurements of one pupil. Repeated patterns in their deviations were deliberately chosen and cannot establish assumptions about a population.

The following code creates the complete data. The factor levels specify the display order and the reference group for subsequent model coefficients. `subset()` selects the twelve pupils assigned to Massed practice; the selected rows still represent twelve distinct pupils.

```
feedback_data <- data.frame(
  student = 1:24,
  feedback = factor(rep(c("Rereading", "Answer feedback",
    "Explanatory feedback"), each = 8),
  levels = c("Rereading", "Answer feedback",
    "Explanatory feedback")),
  practice = factor(rep(rep(c("Massed", "Spaced"),
    each = 4), 3), levels = c("Massed", "Spaced")),
  score = c(4,5,7,8, 4,5,7,8, 6,7,9,10, 10,11,13,14,
    8,9,11,12, 16,17,19,20))
massed <- subset(feedback_data, practice == "Massed")
table(massed$feedback)
#>
#>          Rereading      Answer feedback Explanatory feedback
#>                4                4                4
with(massed, tapply(score, feedback, mean))
#>          Rereading      Answer feedback Explanatory feedback
#>                6                8                10
with(massed, tapply(score, feedback, sd))
#>          Rereading      Answer feedback Explanatory feedback
#>          1.825741858          1.825741858          1.825741858
```

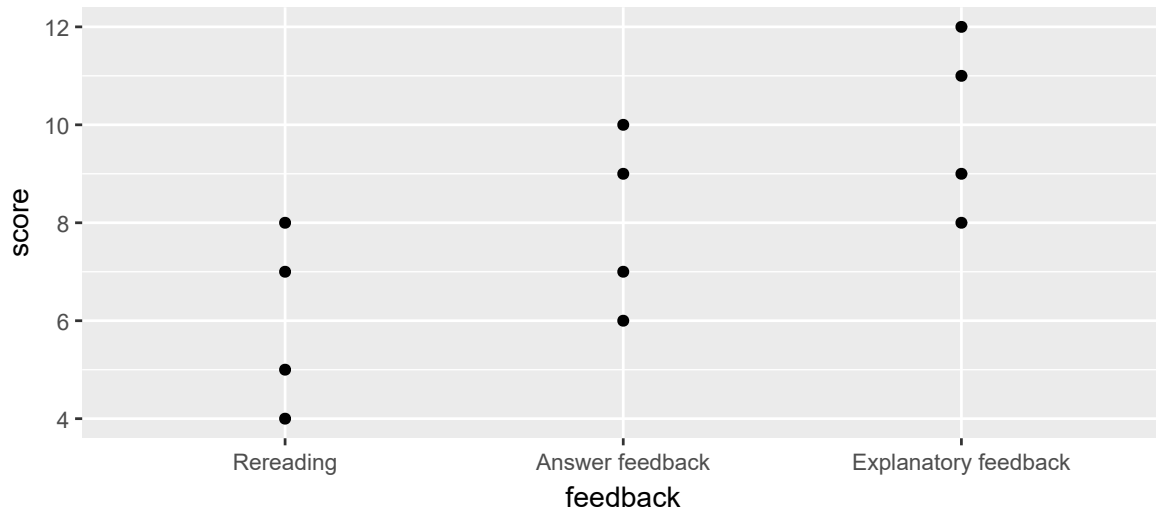
The Massed means are 6, 8 and 10 correct answers, with four pupils in each group and a sample SD of 1.8257 in every group. Their grand mean is 8. Before asking whether a difference is large relative to its uncertainty, we can see the difference on the outcome's own scale: Answer feedback exceeds Rereading by 2 points and Explanatory feedback exceeds it by 4 points.

1.2 Extend a graph one useful component at a time

Chapter 5 used base-R plotting functions. The package `ggplot2` provides another way to describe a graph: name a data frame, map its variables to visual properties, and add the kinds of marks that should be drawn. Its incremental advantage is that another layer or mapping can

extend the same description without rebuilding the graph from separate drawing commands. Load this one package after its installation has been checked in the Computer Lab.

```
library(ggplot2)
p <- ggplot(massed, aes(feedback, score))
p + geom_point()
```



`ggplot(massed, ...)` supplies the records. Inside `aes()`, the `feedback` variable maps to horizontal position and `score` maps to vertical position. `geom_point()` adds one point for each record. The object `p` stores the data and mapping; the expression with `+` adds a plotting component.

We can now add the three group means as larger diamonds. `aggregate()` uses the same twelve scores to make one mean per feedback group. The second point layer inherits the mapping from `p` but uses this three-row data frame.

```
group_means <- aggregate(score ~ feedback, massed, mean)
p + geom_point(size = 2) +
  geom_point(data = group_means, shape = 18, size = 4,
    colour = "black") +
  labs(x = "Feedback", y = "Number correct (out of 20)") +
  theme_minimal(base_size = 11)
```

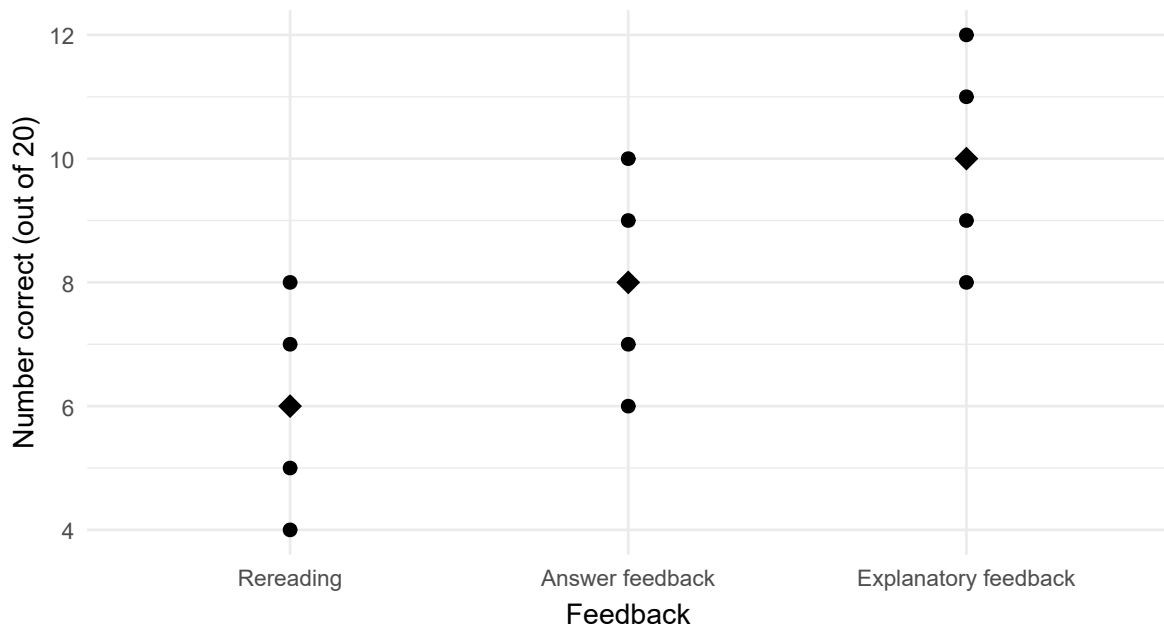


Figure 1: Individual Massed scores and their three group means.

The fixed colour is an argument outside `aes()`. Later, placing a variable inside `aes(colour = ...)` will give different colours to its levels and create a legend. `labs()` names the axes, and `theme_minimal()` changes their presentation. Neither changes the observations or the means. In Figure 1 the raw points keep the denominator and within-group variation visible beside the means.

1.3 Fit the group model and read its decomposition

For pupil i in feedback group j , the one-way model is

$$Y_{ij} = \mu_j + \varepsilon_{ij}.$$

Here μ_j is the group's process mean and ε_{ij} is its pupil's deviation from that mean. Fitting by least squares estimates each μ_j with the corresponding sample mean, so $\widehat{Y}_{ij} = \bar{Y}_j$. The residual is $e_{ij} = y_{ij} - \bar{y}_j$: an observed score minus its fitted group mean. For the first group the fitted values are all 6 and the residuals are $-2, -1, 1, 2$.

The model lets us separate differences between groups from differences among pupils within a group. Chapter 4's decomposition becomes

$$SS_T = \sum_j \sum_i (y_{ij} - \bar{y})^2 = SS_B + SS_W,$$

$$SS_B = \sum_j n_j (\bar{y}_j - \bar{y})^2,$$

$$SS_W = \sum_j \sum_i (y_{ij} - \bar{y}_j)^2.$$

The total measures deviation from the grand mean. The between-group term measures deviation of the three group means from that grand mean, weighted by their group counts. The within-group term measures deviation of pupils from their own fitted group means. In this example, $SS_B = 4(6 - 8)^2 + 4(8 - 8)^2 + 4(10 - 8)^2 = 32$ and each group contributes $(-2)^2 + (-1)^2 + 1^2 + 2^2 = 10$ to SS_W , giving $SS_W = 30$.

The formula `score ~ feedback` tells `aov()` to explain `score` using the `feedback` factor. `anova()` extracts the decomposition from the fitted object; `fitted()` and `resid()` expose its predicted values and residuals.

```
one_fit <- aov(score ~ feedback, data = massed)
anova(one_fit)
#> Analysis of Variance Table
#>
#> Response: score
#>      Df Sum Sq    Mean Sq F value    Pr(>F)
#> feedback    2      32 16.0000000    4.8 0.038131 *
#> Residuals    9      30  3.3333333
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
fitted(one_fit)
#>  1  2  3  4  9 10 11 12 17 18 19 20
#>  6  6  6  6  8  8  8  8 10 10 10 10
resid(one_fit)
#>  1  2  3  4  9 10 11 12 17 18 19 20
#> -2 -1  1  2 -2 -1  1  2 -2 -1  1  2
```

Three fitted group means use three parameters. Consequently the within-group degrees of freedom are $12 - 3 = 9$. The between-group component has $3 - 1 = 2$ degrees of freedom, and the total has $12 - 1 = 11$. Dividing each sum of squares by its degrees of freedom gives a mean square. The within-group mean square $MS_W = 30/9 = 3.3333$ estimates the common error variance.

Table 2*One-way ANOVA on the twelve Massed pupils.*

Source	SS	df	MS	F	p
Feedback	32.0000	2	16.0000	4.8000	.0381
Within groups	30.0000	9	3.3333		
Total	62.0000	11			

The null hypothesis is $H_0 : \mu_R = \mu_A = \mu_E$. Under independent Normal errors with a common variance, the ratio

$$F = \frac{MS_B}{MS_W} = \frac{16}{30/9} = 4.8000$$

has an $F_{2,9}$ reference distribution when that null holds. The p-value .0381 is the probability, under this null model, of an F statistic at least this large. At a .05 threshold it provides evidence against equal process means. It does not say that every pair differs, that all feedback improves performance, or that the invented result establishes a real intervention effect. The next section asks the individual comparison questions.

The observed share of score variation associated with these group means is

$$\eta^2 = \frac{SS_B}{SS_T} = \frac{32}{62} = .5161.$$

This is an in-sample variation summary, accompanying differences expressed in score points. It is neither the probability that feedback caused a result nor a replacement for a confidence interval. Appendix A gives the related bias-adjusted omega-squared calculation, with its denominator and interpretation.

1.4 Express the same fit with indicators

A factor with three levels can also enter a linear model as two indicator variables. Let $D_A = 1$ for Answer feedback and 0 otherwise, and let $D_E = 1$ for Explanatory feedback and 0 otherwise. Rereading has both indicators 0. The fitted equation is

$$\hat{y} = 6 + 2D_A + 4D_E.$$

Substituting the indicators gives 6 for Rereading, $6 + 2 = 8$ for Answer feedback and $6 + 4 = 10$ for Explanatory feedback. The intercept is the reference mean; each other coefficient is a

difference from that mean. The factor order in the original setup makes Rereading the reference level.

```
indicator_fit <- lm(score ~ feedback, data = massed)
coef(indicator_fit)
#>              (Intercept)      feedbackAnswer feedback
#>                   6              2
#> feedbackExplanatory feedback
#>                   4
anova(indicator_fit)
#> Analysis of Variance Table
#>
#> Response: score
#>      Df Sum Sq   Mean Sq F value    Pr(>F)
#> feedback    2     32 16.000000    4.8 0.038131 *
#> Residuals    9     30  3.333333
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

`lm()` and `aov()` describe the same set of fitted means and give the same residuals and ANOVA here. The formula names the outcome and explanatory variable; the study’s design supplies the independence assumption. This indicator representation will make the adjusted group comparison easier to read. Chapter 7 develops a continuous predictor’s slope, and Chapter 8 extends the conditional interpretation to several predictors.

2 Ask comparisons within stated families

2.1 Specify the comparison and its standard error

Suppose the two questions concern Answer feedback versus Rereading and Explanatory feedback versus Rereading, among Massed pupils. In the order Rereading, Answer feedback, Explanatory feedback, their coefficient vectors are $(-1, 1, 0)$ and $(-1, 0, 1)$. Each vector sums to zero. A quantity

$$C = \sum_j c_j \mu_j, \quad \sum_j c_j = 0,$$

is a contrast of the process means. Its estimate replaces process means with the observed group means: $\widehat{C} = \sum_j c_j \bar{y}_j$. Thus our estimates are 2 and 4 points. Calling a comparison a contrast specifies this mathematical form; it does not say whether the comparison was planned before seeing the results or selected afterward. That is additional information about the analysis.

For independent groups with a common error variance, Chapter 3's variance rule for a linear combination gives

$$SE(\widehat{C}) = \sqrt{MS_W \sum_j \frac{c_j^2}{n_j}}.$$

Each comparison uses two of the means, so both have estimated SE $\sqrt{(30/9)(1/4 + 1/4)} = 1.2910$ points. The third group still contributes to the common error estimate and its 9 residual degrees of freedom. This matters when comparing these results with an analysis restricted to only two groups.

2.2 Calibrate the two-comparison family

An individual 95% interval uses a .05 error probability for that one comparison. To cover both declared comparisons with at least 95% familywise confidence, Bonferroni allocates .025 to each two-sided comparison. Each tail then receives $.05/(2 \times 2) = .0125$. The calibration uses the t reference distribution with the model's 9 residual degrees of freedom (Student, 1908), giving the critical value

$$t_{1-.05/(2 \times 2), 9} = 2.6850.$$

The calculations retain the same estimates and SEs but multiply each SE by this larger critical value. The following code extracts the common variance estimate from the fitted object and calculates both results from unrounded inputs.

```
mse <- deviance(one_fit) / df.residual(one_fit)
means <- with(massed, tapply(score, feedback, mean))
estimate <- c(Answer = unname(means[2] - means[1]),
  Explain = unname(means[3] - means[1]))
se <- sqrt(mse * (1/4 + 1/4))
critical <- qt(1 - .05/(2*2), df.residual(one_fit))
raw_p <- 2 * pt(-abs(estimate/se), df.residual(one_fit))
cbind(estimate, SE = se, lower = estimate-critical*se,
  upper = estimate+critical*se,
  adjusted_p = p.adjust(raw_p, method = "bonferroni"))
#>      estimate      SE      lower      upper
#> Answer      2 1.290994449 -1.466334098  5.466334098
#> Explain      4 1.290994449  0.533665902  7.466334098
#>      adjusted_p
#> Answer 0.31148928496
#> Explain 0.02551124295
```

Table 3*Bonferroni results for the declared family of two comparisons.*

Comparison	Estimate	SE	Simultaneous interval	Adjusted p
Answer minus Rereading	2.0000	1.2910	[−1.4663, 5.4663]	.3115
Explanatory minus Rereading	4.0000	1.2910	[0.5337, 7.4663]	.0255

Multiplying each raw p-value by two and capping at one gives the corresponding Bonferroni-adjusted p-value. Explanatory minus Rereading is distinguishable from zero with this familywise calibration, whereas Answer minus Rereading is not. Its interval allows both a small negative difference and a positive difference of several points. This uncertainty is compatible with the earlier omnibus rejection: the omnibus test and each contrast address different statements.

2.3 Compare with an all-pairs family

If the question includes every pair, there are three comparisons. Tukey’s procedure uses the joint distribution of pairwise mean differences under the common-variance Normal-error model. `TukeyHSD()` supplies those simultaneous intervals and their adjusted p-values (Tukey, 1953).

```
TukeyHSD(one_fit, "feedback", conf.level = .95)
#> Tukey multiple comparisons of means
#> 95% family-wise confidence level
#>
#> Fit: aov(formula = score ~ feedback, data = massed)
#>
#> $feedback
#>
#>      diff      lwr
#> Answer feedback-Rereading      2 -1.6044637455
#> Explanatory feedback-Rereading      4  0.3955362545
#> Explanatory feedback-Answer feedback      2 -1.6044637455
#>
#>      upr      p adj
#> Answer feedback-Rereading      5.604463746 0.3149864972
#> Explanatory feedback-Rereading      7.604463746 0.0310140456
#> Explanatory feedback-Answer feedback      5.604463746 0.3149864972
```

The output labels each subtraction direction. Explanatory minus Rereading is 4, with interval [0.3955, 7.6045] and adjusted p .0310. Answer minus Rereading and Explanatory minus Answer both equal 2, with intervals [−1.6045, 5.6045] and adjusted p .3150. These limits differ from the preceding intervals because the declared family and its calibration differ. An interval should therefore be read together with the comparison and family it represents.

An omnibus rejection does not by itself protect any comparison selected after examining the means. Conversely, a separately specified contrast can have its own valid uncertainty procedure without requiring omnibus rejection as an automatic preliminary gate. The relevant information is the question, selection process and calibration, rather than a universal sequence of significance decisions.

2.4 Check what changes when the variance model changes

The common-variance model assumes that within-group variability estimates one shared error variance. In a real study, design information and observed patterns help assess that working model. A small normality or variance test cannot certify it, especially with four pupils per group. Here the identical SDs were deliberately constructed.

For independent groups whose error variances may differ, Welch’s one-way procedure provides an alternative (Welch, 1947). R selects it by default in `oneway.test()` (R Core Team, 2025):

```
oneway.test(score ~ feedback, data = massed)
#>
#> One-way analysis of means (not assuming equal variances)
#>
#> data:  score and feedback
#> F = 4.32, num df = 2, denom df = 6, p-value = 0.06883836
```

It gives $F(2,6) = 4.3200$, $p .0688$. Although the observed group SDs happen to be equal, Welch’s small-sample correction and reference degrees of freedom differ from the pooled-variance ANOVA. Its uncertainty calculation is built for a different variance model. The two p-values are therefore results to explain through their assumptions and calibration, rather than votes on which numerical conclusion to prefer. With small groups, the model choice can matter appreciably.

A useful exact identity connects a two-group pooled t test with a two-group ANOVA: $F = t^2$, with the same two-sided p-value. It requires the same records, equal-variance denominator and group comparison. R’s default Welch t test does not meet that specification. Nor is our three-group contrast the same calculation, because it uses residual information from the third group. Appendix A reproduces the matched identity explicitly.

3 Compare a two-way pattern

3.1 Read cells and margins before fitting

We now return to all 24 pupils and ask how the Spaced-minus-Massed comparison changes across feedback conditions. A cell is one combination of feedback and practice. Computing cell counts and means first makes the six cells and their denominators visible.

```
with(feedback_data, table(feedback, practice))
#>               practice
#> feedback      Massed Spaced
#> Rereading           4      4
#> Answer feedback      4      4
#> Explanatory feedback  4      4
cell_mean <- with(feedback_data,
  tapply(score, list(feedback, practice), mean))
cell_mean
#>               Massed Spaced
#> Rereading           6      6
#> Answer feedback      8     12
#> Explanatory feedback 10     18
rowMeans(cell_mean)
#>      Rereading      Answer feedback Explanatory feedback
#>              6              10              14
colMeans(cell_mean)
#> Massed Spaced
#>      8     12
mean(feedback_data$score)
#> [1] 10
```

Table 4

Cell means and equally weighted margins of the balanced design.

Feedback	Massed	Spaced	Equally weighted margin	Spaced minus Massed
Rereading	6	6	6	0
Answer feedback	8	12	10	4
Explanatory feedback	10	18	14	8
Practice margin	8	12	10	4

Each cell has four pupils, so averaging the two practice cells equally gives the feedback group's overall mean. Similarly, each practice margin averages equally across the three feedback

conditions. In an unbalanced design, a margin must specify its weights: the observed mix of pupils and an equally weighted target can answer different questions.

The new graph uses the same data/mapping/layer idea as the earlier group plot. Here a colour identifies each feedback condition and **group** identifies the means to connect. The line joins separate cell means; it does not trace individual pupils across repeated measurements.

```
cell_plot <- aggregate(score ~ feedback + practice,
  feedback_data, mean)
ggplot(cell_plot, aes(practice, score, colour = feedback,
  group = feedback)) + geom_line() + geom_point(size = 2) +
  labs(x = "Practice", y = "Mean number correct",
    colour = "Feedback") +
  theme_minimal(base_size = 11)
```

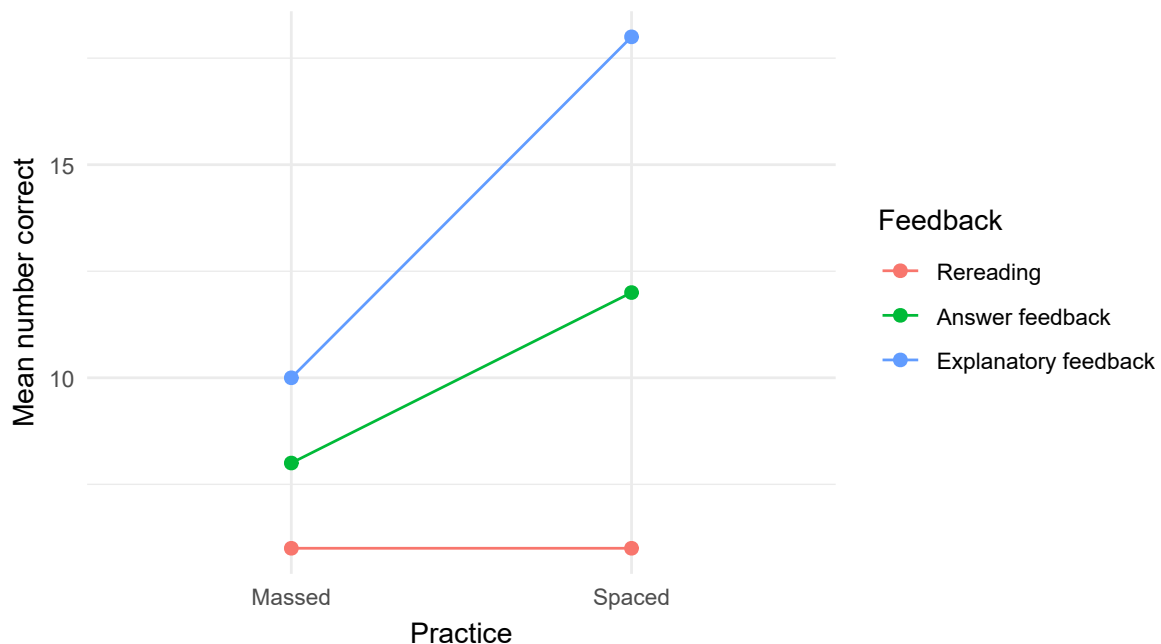


Figure 2: The practice difference depends on feedback.

`colour = feedback` is inside `aes()` because a variable now determines colour. The mapping also creates the legend. The observed changes are 0, 4 and 8 points, producing nonparallel lines in Figure 2. This describes interaction in the observed means. Inference about a process interaction additionally requires a model and a measure of residual variation.

3.2 Fit the full model and identify its tests

Chapter 4 wrote the two-way model as

$$Y_{ijk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} + \varepsilon_{ijk},$$

where j indexes feedback, k indexes practice and i indexes pupils within a cell. With the usual sum-to-zero constraints, μ is the grand mean, α_j and β_k describe departures of the margins from it, and $(\alpha\beta)_{jk}$ describes the remaining cell departure. The constraints make this representation identifiable: the effects cannot all shift freely while leaving the same fitted means.

There is one grand-mean parameter, $3 - 1 = 2$ feedback degrees of freedom, $2 - 1 = 1$ practice degree of freedom, and $(3-1)(2-1) = 2$ interaction degrees of freedom. Together these describe six fitted cell means. The model therefore has $24 - 6 = 18$ residual degrees of freedom.

In an R formula, `feedback * practice` includes both main effects and their interaction. It expands to `feedback + practice + feedback:practice`.

```
two_fit <- aov(score ~ feedback * practice,
  data = feedback_data)
anova(two_fit)
#> Analysis of Variance Table
#>
#> Response: score
#>
#>           Df Sum Sq   Mean Sq F value    Pr(>F)
#> feedback      2    256 128.000000    38.4 3.2076e-07 ***
#> practice      1     96  96.000000    28.8 4.2288e-05 ***
#> feedback:practice  2     64  32.000000     9.6  0.001454 **
#> Residuals    18     60   3.333333
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 5

Full balanced factorial decomposition.

Source	SS	df	MS	F	p
Feedback	256.0000	2	128.0000	38.4000	< .0001
Practice	96.0000	1	96.0000	28.8000	< .0001
Interaction	64.0000	2	32.0000	9.6000	.0015
Residual	60.0000	18	3.3333		
Total	476.0000	23			

The feedback test concerns equality of feedback margins averaged equally over practice. The practice test concerns equality of practice margins averaged equally over feedback. The interaction test asks whether the practice difference is the same across all feedback conditions. With independent Normal errors of common variance, its null reference is $F_{2,18}$; the observed value 9.6000 and p .0015 provide evidence against that common-difference null.

The balanced layout makes the source components orthogonal: the variation attributed to one component does not compete with another component for the same fitted direction. A single-model `anova()` table ordinarily uses sequential sums of squares, entering terms in formula order. This convenient balanced decomposition should not be extended automatically to unbalanced or adjusted data. Later in this chapter, we compare explicitly named nested models on identical rows to specify adjusted questions.

The feedback and practice margins remain meaningful descriptions, but the interaction clarifies their limits. For example, the 4-point overall practice difference averages a 0-point, a 4-point and an 8-point difference. A result paragraph should convey this varying pattern when the scientific question concerns how practice works across feedback conditions.

3.3 Work the selected difference of differences completely

Consider Answer feedback's practice change minus Rereading's practice change:

$$\widehat{DD} = (12 - 8) - (6 - 6) = 4.$$

The contrast has coefficients $(1, -1, -1, 1)$ in the cell order Rereading–Massed, Rereading–Spaced, Answer–Massed, Answer–Spaced. Its four cell means come from different pupils. Under the common-variance model, their variances therefore add after multiplying by the squared coefficients:

$$SE(\widehat{DD}) = \sqrt{MS_E \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right)} = 1.8257.$$

The omnibus interaction spans two independent directions. For simultaneous confidence over all contrasts in that two-dimensional interaction space, Scheffe's method uses the multiplier

$$\sqrt{2F_{.95;2,18}} = 2.6663.$$

The complete calculation keeps the cell subtraction, residual variance and family calibration visible:

```

dd <- (cell_mean["Answer feedback", "Spaced"] -
      cell_mean["Answer feedback", "Massed"]) -
      (cell_mean["Rereading", "Spaced"] -
      cell_mean["Rereading", "Massed"])
mse_two <- deviance(two_fit) / df.residual(two_fit)
se_dd <- sqrt(mse_two * 4/4)
scheffe <- sqrt(2 * qf(.95, 2, 18))
c(estimate = dd, SE = se_dd, multiplier = scheffe)
#>      estimate      SE multiplier
#> 4.0000000000 1.825741858 2.666292237
dd + c(-1, 1) * scheffe * se_dd
#> [1] -0.8679613431  8.8679613431

```

The resulting interval is $4 \pm 2.6663(1.8257) = [-0.8680, 8.8680]$. It includes zero even though the interaction F test rejects its null. The F test asks whether both interaction directions are zero jointly; this interval addresses one selected direction while allowing for the entire family. Other directions can carry substantial evidence, and a simultaneous interval for this direction can remain broad. Neither result cancels the other.

Three related quantities help distinguish the questions. Answer feedback's simple practice effect is 4, with SE 1.2910 and pointwise t interval [1.2877, 6.7123]. The selected DD also happens to equal 4, but its SE is 1.8257 because it compares two practice effects. Its pointwise interval is [0.1643, 7.8357], whereas its simultaneous interval is [-0.8680, 8.8680]. Appendix A shows both DD calculations using the same estimate and SE.

A simple effect includes the overall practice component as well as an interaction component. It is therefore outside the interaction-only family used above. The shared numerical estimate does not give it the DD's meaning, SE or familywise guarantee. State which effect was selected, which family was considered and which interval was used before interpreting a result.

4 Adjust a group comparison for baseline

4.1 State the new data and the conditional question

Suppose we want to compare group means at a common pretest score. The earlier 24-pupil dataset has no pretest. We therefore introduce a separate authored example of twelve pupils in Table 6. These are new teaching inputs, with individual assignment to feedback stipulated to occur after the pretest. Each row records one pupil's two assessments, both counting correct answers out of 20. The pupil is the assignment and measurement unit; distinct pupils supply the independent error units in the working model.

Table 6*Fixed inputs for a separate baseline-adjustment illustration.*

Pupil	Feedback	Pretest	Posttest
1	Rereading	4	5
2	Rereading	6	9
3	Rereading	8	11
4	Rereading	10	11
5	Answer feedback	6	9
6	Answer feedback	8	13
7	Answer feedback	10	15
8	Answer feedback	12	15
9	Explanatory feedback	5	10
10	Explanatory feedback	7	14
11	Explanatory feedback	9	16
12	Explanatory feedback	11	16

Missing values are absent by construction. The regular residual pattern makes adjustment arithmetic easy to follow, but cannot demonstrate Normal errors, independence or a common slope in a real study. The observed pretest ranges are 4–10, 6–12 and 5–11, so their common range is 6–10. Pretest 8 lies within each group’s observed range. Comparing fitted means there is more supportable than extending a group’s line far beyond its observations.

```
feedback_adjustment <- data.frame(
  student = 1:12,
  feedback = factor(rep(c("Rereading", "Answer feedback",
    "Explanatory feedback"), each = 4),
    levels = c("Rereading", "Answer feedback",
      "Explanatory feedback")),
  pretest = c(4,6,8,10, 6,8,10,12, 5,7,9,11),
  posttest = c(5,9,11,11, 9,13,15,15, 10,14,16,16))
feedback_adjustment$baseline <- feedback_adjustment$pretest - 8
aggregate(cbind(pretest, posttest) ~ feedback,
  feedback_adjustment, mean)
#>           feedback pretest posttest
#> 1      Rereading         7         9
#> 2 Answer feedback         9        13
#> 3 Explanatory feedback      8        14
```

The raw posttest means are 9, 13 and 14, while the pretest means are 7, 9 and 8. A raw group mean averages over that group’s own baselines. Our new question fixes pretest at 8 and

compares the fitted posttest means there. The difference between these questions explains why the adjusted differences can differ from the raw differences.

4.2 Describe and fit a common-slope model

Analysis of covariance, or ANCOVA, uses a linear model that includes both group indicators and quantitative covariates. Define the centered baseline $x_c = x - 8$. With the same indicator meanings as before, the model is

$$Y_i = \beta_0 + \beta_x x_{ci} + \beta_A D_{Ai} + \beta_E D_{Ei} + \varepsilon_i.$$

At $x = 8$, the centered baseline is zero. The intercept is therefore Rereading's fitted mean at pretest 8. The two group coefficients are differences from Rereading at that same pretest. The common slope β_x describes the fitted change in mean posttest for a one-point pretest change within any group. Graphically, the model uses three parallel lines.

Centering changes the intercept's reference point, while leaving the fitted values unchanged. Choosing 8 also makes the intercept refer to a value supported by observations in all three groups. Chapter 7 develops how a slope is fitted, and Chapter 8 develops conditioning on several predictors; here the line gives a concrete first adjusted comparison.

```
unadjusted <- lm(posttest ~ feedback,
  data = feedback_adjustment)
adjusted <- lm(posttest ~ baseline + feedback,
  data = feedback_adjustment)
coef(summary(adjusted))
#>
#> Estimate Std. Error t value
#> (Intercept) 10 0.6324555320 15.811388301
#> baseline 1 0.1581138830 6.324555320
#> feedbackAnswer feedback 2 0.9219544457 2.169304578
#> feedbackExplanatory feedback 4 0.8803408431 4.543694674
#>
#> Pr(>|t|)
#> (Intercept) 2.559753259e-07
#> baseline 2.267167452e-04
#> feedbackAnswer feedback 6.188555919e-02
#> feedbackExplanatory feedback 1.889983199e-03
confint(adjusted)
#>
#> 2.5 % 97.5 %
#> (Intercept) 8.5415549278 11.458445072
#> baseline 0.6353887319 1.364611268
#> feedbackAnswer feedback -0.1260307643 4.126030764
#> feedbackExplanatory feedback 1.9699303755 6.030069625
```

Table 7*Common-slope fit, with residual MSE 1.5000 on 8 df.*

Term	Estimate	SE	95% pointwise interval
Rereading at pretest 8	10.0000	0.6325	[8.5416, 11.4584]
Pretest slope	1.0000	0.1581	[0.6354, 1.3646]
Answer minus Rereading	2.0000	0.9220	[−0.1260, 4.1260]
Explanatory minus Rereading	4.0000	0.8803	[1.9699, 6.0301]

`coef(summary(adjusted))` returns estimates, SEs, t statistics and two-sided p-values for the individual zero-coefficient tests. The adjusted Answer difference has p .0619 and the Explanatory difference has p .0019. `confint()` uses the residual t reference to form the corresponding intervals. These intervals are pointwise: a simultaneous conclusion about a family requires an explicitly chosen family adjustment.

4.3 Read fitted means at the same baseline

To obtain a fitted mean for each group at pretest 8, construct one profile per group with `baseline = 0`. `predict()` evaluates the fitted equation at each profile and returns an interval for its conditional mean.

```
profiles <- data.frame(baseline = 0,
  feedback = levels(feedback_adjustment$feedback))
predict(adjusted, newdata = profiles,
  interval = "confidence")
#>   fit      lwr      upr
#> 1  10  8.541554928 11.45844507
#> 2  12 10.541554928 13.45844507
#> 3  14 12.587866631 15.41213337
```

The adjusted means are 10, 12 and 14. Their 95% pointwise confidence intervals are [8.5416, 11.4584], [10.5416, 13.4584] and [12.5879, 15.4121], respectively. They estimate group means at a stated baseline, rather than ranges containing individual future pupils. Prediction for a new pupil would include that pupil's additional residual variation; Chapter 8 develops the distinction.

The adjustment can also be read directly from the raw means. For the common-slope fit,

$$\hat{\mu}_j(8) = \bar{y}_j - \hat{\beta}_x(\bar{x}_j - 8).$$

With slope 1, Rereading's adjusted mean is $9 - (7 - 8) = 10$; Answer feedback's is $13 - (9 - 8) = 12$; Explanatory feedback's is $14 - (8 - 8) = 14$. Thus the raw differences of 4 and 5 points become conditional differences of 2 and 4 points. Adjustment compares groups at a common baseline rather than at their different average baselines.

Figure 3 shows this geometry. The fitted segments are restricted to the shared observed range 6–10, and the dashed line marks pretest 8. The points still show the full range of each group's data.

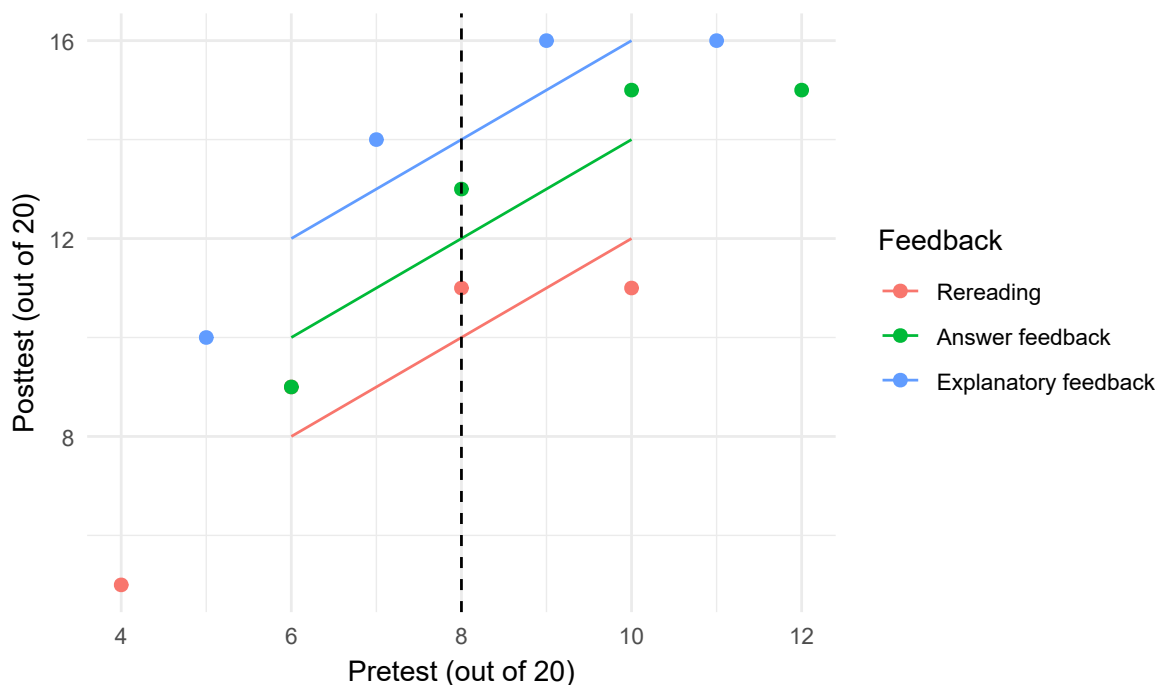


Figure 3: Compare fitted group means at a shared baseline.

The Computer Lab's adjustment reference constructs the plotting grid and changes the supported baseline value. The statistical calculation remains the printed `predict()` call: specify the profiles first, then interpret the fitted means at those profiles.

4.4 Compare named nested models on identical records

Two questions about this fit require different smaller models. First, does baseline add information after accounting for the feedback groups? Compare the groups-only fit, `unadjusted`, with the larger fit, `adjusted`. Second, do the group indicators jointly add information after accounting for baseline? Compare a baseline-only fit with the same larger fit.

```

baseline_only <- lm(posttest ~ baseline,
  data = feedback_adjustment)
anova(unadjusted, adjusted)
#> Analysis of Variance Table
#>
#> Model 1: posttest ~ feedback
#> Model 2: posttest ~ baseline + feedback
#>   Res.Df  RSS Df Sum of Sq  F      Pr(>F)
#> 1      9   72
#> 2      8  12  1      60 40 0.00022672 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
anova(baseline_only, adjusted)
#> Analysis of Variance Table
#>
#> Model 1: posttest ~ baseline
#> Model 2: posttest ~ baseline + feedback
#>   Res.Df      RSS Df Sum of Sq      F      Pr(>F)
#> 1      10 43.058824
#> 2      8 12.000000  2 31.058823 10.35294 0.0060322 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

For nested models fitted to the same outcome and rows, let RSS_0 and df_0 denote the smaller model's residual sum of squares and degrees of freedom, and let RSS_1 and df_1 denote the larger model's. Then

$$F = \frac{(RSS_0 - RSS_1)/(df_0 - df_1)}{RSS_1/df_1}.$$

The numerator measures the improvement per added parameter. The denominator is the residual variance estimate from the larger model. Here the larger model estimates four coefficients from twelve pupils, leaving 8 residual degrees of freedom and $RSS_1/df_1 = 12/8 = 1.5000$.

Adding baseline to groups reduces RSS from 72 on 9 df to 12 on 8 df, giving $F(1, 8) = 40.0000$, $p .0002$. Adding the group indicators to baseline alone reduces RSS from 43.0588 on 10 df to 12 on 8 df, giving $F(2, 8) = 10.3529$, $p .0060$. The latter tests both adjusted group coefficients jointly. As with the earlier omnibus tests, it does not establish that each individual adjusted comparison differs from zero.

The same-row condition is essential. If different predictors contain missing values, automatic omission can fit the smaller and larger models to different pupils. Their RSS difference would then mix a model change with a data change. Explicit model comparison also clarifies the

question more reliably than treating every entry in a single sequential ANOVA table as an order-free adjusted test.

4.5 Check the fit and state its scope

The classical tests and intervals use independent errors, an appropriate linear common-slope conditional mean, a common conditional error variance, and a Normal-error reference. A graph of the fitted relationship helps locate unsupported extrapolation or a changing group comparison. A residual plot examines departures left after fitting; a Normal-quantile plot compares their shape with Normal quantiles.

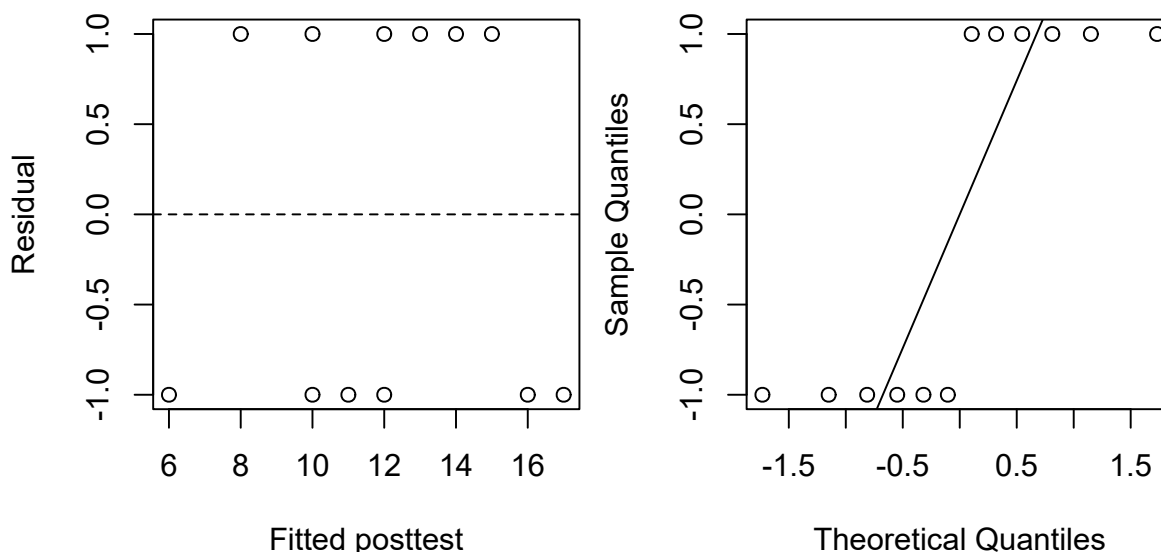


Figure 4: Residual checks describe the deliberately regular ANCOVA construction.

In Figure 4, residuals near -1 and 1 reflect the deliberately regular construction. The display illustrates what to examine; these twelve chosen values cannot validate a Normal population model. Similarly, a nonsignificant test for a group-by-baseline term would not prove equal slopes. Such a term asks whether the adjusted group comparison changes with baseline; later chapters develop that moderation question.

In real data, report the observed denominators and exclusions as well as the fitted model. Random allocation does not restore unavailable outcomes. Adjustment alone does not identify a causal effect in observational data, repair missing outcomes, or change the population sampled. Here the assignment story and values are invented; the result demonstrates an analysis, not an intervention finding. Chapter 8 develops several controls, covariance-based fitted quantities and comparisons averaged over a named population. Chapter 9 carries the separate Baumann adjusted-mean reference as an independent extension.

5 Analyse proportions and counts

5.1 Keep the event and denominator visible

A score model is not the appropriate starting point for every outcome. Chapter 3 introduced the invented completion table in Table 8. It contains aggregate counts for 100 records, classified by learning format and completion. These records have no pupil-level linkage to either feedback dataset above. The counts themselves establish neither random allocation nor representative sampling.

Table 8

The same authored completion counts as Chapter 3.

Format	Completed	Not completed	Total
Online	32	18	50
In person	18	32	50
Total	50	50	100

For the Online group, the event is completion and the denominator is its 50 records. Under an independent Bernoulli model with a common completion probability p , the count of completions has a binomial distribution. The estimated probability is $\hat{p} = 32/50 = .6400$. If records were clustered within dependent classes or came from repeated observations of the same people, this ordinary independent-unit uncertainty model would require modification.

Suppose the benchmark is $p_0 = .5$. The score test standardizes the difference using the variance under that null:

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/50}} = \frac{.64 - .5}{\sqrt{.5(.5)/50}} = 1.9799.$$

Squaring gives $X^2 = z^2 = 3.9200$, compared with a chi-squared distribution on one degree of freedom. The procedure uses a large-sample reference; the expected numbers of successes and failures under this null are both 25.

```

one_prop <- prop.test(32, 50, p = .5, correct = FALSE)
one_prop$estimate
#>      p
#> 0.64
one_prop$statistic
#> X-squared
#>      3.92
one_prop$p.value
#> [1] 0.04771488024
one_prop$conf.int
#> [1] 0.5014101688 0.7586125035
#> attr(,"conf.level")
#> [1] 0.95

```

The two-sided p-value is .0477. The estimated completion probability is .6400, with 95% Wilson score interval [.5014, .7586]. `correct = FALSE` explicitly selects the uncorrected score procedure; accepting R's default continuity correction would specify a different calculation. The call and the reported result should identify which procedure was used.

The score interval contains the candidate probabilities that their own two-sided score tests would retain (Wilson, 1927). Each candidate supplies the variance for its null calculation. This inversion explains why the interval is linked to the score test and is not simply $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$, the Wald interval. Appendix A gives the Wilson endpoints directly. The p-value assesses the stipulated benchmark; it is not the probability that $p = .5$ is true or a measure of the completion rate's practical importance.

5.2 Compare the two groups through their full table

Now ask whether completion probability differs by format. This uses both rows and a different null: format and completion are independent in the working population model, equivalently equal completion probabilities in this two-group setting. Constructing labelled counts prevents an accidental change in the event or group direction.

```

counts <- matrix(c(32, 18, 18, 32), nrow = 2,
  byrow = TRUE,
  dimnames = list(Format = c("Online", "In person"),
    Result = c("Completed", "Not completed")))
counts
#>           Result
#> Format      Completed Not completed
#>   Online           32           18
#> In person          18           32
prop.table(counts, margin = 1)
#>           Result
#> Format      Completed Not completed
#>   Online           0.64           0.36
#> In person          0.36           0.64
count_test <- chisq.test(counts, correct = FALSE)
count_test$expected
#>           Result
#> Format      Completed Not completed
#>   Online           25           25
#> In person          25           25
count_test$statistic
#> X-squared
#>           7.84
count_test$p.value
#> [1] 0.005110260661
count_test$residuals
#>           Result
#> Format      Completed Not completed
#>   Online           1.4          -1.4
#> In person          -1.4           1.4

```

`margin = 1` divides each row by its own total. Completion rates are .6400 for Online and .3600 for In person, a difference of .2800, or 28.0000 percentage points, in the Online-minus-In-person direction. Under independence the fitted expected count in row j , column k is

$$E_{jk} = \frac{(\text{row } j \text{ total})(\text{column } k \text{ total})}{N}.$$

All four expected counts here equal $50 \times 50/100 = 25$. Pearson's statistic accumulates the squared discrepancies relative to those expected counts:

$$X^2 = \sum_{j,k} \frac{(O_{jk} - E_{jk})^2}{E_{jk}} = 4 \left(\frac{7^2}{25} \right) = 7.8400.$$

For a J -by- K table the usual independence reference has $(J - 1)(K - 1)$ degrees of freedom. Here $df = 1$ and $p = .0051$. The observed Pearson statistic X^2 is a number calculated from the table; χ^2_1 names its approximate reference distribution under the null model.

The Pearson residuals are $(O - E)/\sqrt{E}$: 1.4, -1.4, -1.4, 1.4 in row order. Their signs locate the pattern: more Online completions and fewer Online non-completions than the independence fit predicts, with the opposite departures for In person. Squaring and adding these residuals recovers X^2 . They are related through the table margins, so their signs do not constitute four independent cell tests.

The approximation concerns expected counts under the fitted null model, along with independent observational units and a sampling model appropriate to the table. Looking only at observed cell counts misses part of this check. Sparse expected cells may require an alternative reference or a more careful model. In the present example all expected counts are 25, supporting the ordinary chi-squared approximation under the stated unit model.

The one-proportion test asked whether Online completion probability equals a fixed .5 benchmark. The complete-table test asks whether the two format probabilities are equal; it estimates their common probability under the null from the full table. Their different p-values answer different questions. Neither test establishes that changing format would cause completion to change without relevant design information.

5.3 Use alternatives as changes to a stated analysis

The Computer Lab's categorical reference compares the two-proportion interface, a continuity-corrected calculation, Fisher's conditional test (Fisher, 1935) and a standardized association summary, Cramer's V (Cramér, 1946), using the same table. These alternatives keep their own conditions and uncertainty calculations. For example, the two-proportion difference interval is a large-sample interval for a difference, while the preceding Wilson interval concerns one probability. A shared software function name does not make the intervals identical procedures.

That reference also continues Chapters 2–3's observational student intention-by-pass example. Its complete primary calculation keeps zero grades, states the original denominators and compares the outcome under a documented source-coding sensitivity analysis. This is a separate application with its own provenance. The small invented completion table above supplies the first complete procedure; the reference shows how method choice and unresolved source information matter in an existing dataset.

6 Fit by an objective function and likelihood

6.1 Fix the data and vary a candidate parameter

The fitted means in Section 1 were chosen by a rule: minimize the sum of squared deviations from the observations. We can inspect this rule with the four Massed Rereading scores, 4, 5, 7 and 8. These observations stay fixed. What changes is the proposed center, denoted by c .

An **objective function** assigns a numerical criterion to each candidate parameter value for fixed data. For a center fitted by least squares, the objective is

$$Q(c) = \sum_{i=1}^4 (y_i - c)^2 = (4 - c)^2 + (5 - c)^2 + (7 - c)^2 + (8 - c)^2.$$

At $c = 6$, the squared deviations sum to $4 + 1 + 1 + 4 = 10.0000$ squared score points. At either 5 or 7, they sum to 14.0000. Chapter 2 showed why the sample mean minimizes this criterion. Here the mean is 6.0000 points. A numerical search lets us reproduce the minimizing value without writing it into the fitting instruction.

```
rereading <- subset(massed, feedback == "Rereading")$score
loss <- function(center) sum((rereading - center)^2)
mean_fit <- optimize(loss, interval = range(rereading))
c(center = mean_fit$minimum, squared_loss = mean_fit$objective)
#>      center squared_loss
#>          6          10
c(loss_at_5 = loss(5), loss_at_6 = loss(6), loss_at_7 = loss(7))
#> loss_at_5 loss_at_6 loss_at_7
#>      14      10      14
```

`function(center)` makes an R function whose input is a candidate center. Subtracting it from the vector forms four deviations; squaring and summing returns one number. `optimize()` searches for a minimum of that function over the interval supplied. Here `range(rereading)` gives $[4, 8]$, which contains the sample mean. The objective is a single smooth bowl over this interval, so the scalar search has one minimum to locate.

The two returned quantities have different meanings. `minimum` is the **parameter location**, 6.0000 points; `objective` is the function value there, 10.0000 squared points. The latter is a sum of squared deviations, not an SE or an estimate of the mean. The search interval also belongs to the fitting instruction: excluding the best center would restrict which answer the search could return. Numerical stopping tolerance concerns the accuracy of that search; statistical uncertainty concerns what an estimate would do across the repetitions justified by the study.

These least-squares identities are arithmetic. An uncertainty statement for the fitted mean adds a model or design and its conditions. Chapter 7 extends this criterion from one center to a line with an intercept and slope. The Computer Lab reference varies the search interval while keeping the observations fixed.

6.2 Change the fitting criterion for a binomial outcome

For the completion example, the observations are 32 completions in 50 Online trials. Let p denote a candidate probability of completion. Under the stipulated model of independent trials with a common probability, the number completed follows a binomial distribution. The model assigns probability

$$P_p(X = 32) = \binom{50}{32} p^{32} (1 - p)^{18}.$$

In a probability calculation, we choose a value of p and consider which counts might occur. For fitting, we have already observed the count: 32 stays fixed, as does the total of 50. We now evaluate the same expression at different candidate values of p . Viewed in this direction, it is the **likelihood function**,

$$L(p) = \binom{50}{32} p^{32} (1 - p)^{18}, \quad 0 \leq p \leq 1.$$

The binomial coefficient counts the ways 32 completions can occur among 50 trials. It is constant as p varies. In this binomial model, likelihood is the probability of the observed data under each candidate model parameter; it is not a probability distribution over that parameter. In particular, a high likelihood at a candidate p does not give the probability that p is true.

Maximum likelihood estimation chooses the parameter value that makes the observed data most likely under the declared model. The logarithm is increasing, so maximizing the log-likelihood gives the same parameter value. For $0 < p < 1$,

$$\ell(p) = \log \binom{50}{32} + 32 \log p + 18 \log(1 - p).$$

Logs turn a product of powers into a sum and make calculation easier. The binomial coefficient affects the reported function value, although it cannot change the maximizing p . Here the interior maximum can also be found analytically. Differentiating with respect to the candidate probability gives

$$\ell'(p) = \frac{32}{p} - \frac{18}{1-p} = \frac{32-50p}{p(1-p)}.$$

Setting this slope to zero gives $32-50p=0$, and therefore $\hat{p}=32/50=.6400$. The denominator is positive inside $(0,1)$. The slope is positive below .64 and negative above .64: the curve rises to this point and then falls. This verifies a maximum rather than merely finding a flat point.

To use the default minimization operation in `optimize()`, supply the **negative** log-likelihood. A smaller negative log-likelihood means a larger log-likelihood, so its minimum locates the same fitted probability.

```
negative_loglik <- function(p) {
  -dbinom(32, size = 50, prob = p, log = TRUE)
}
likelihood_fit <- optimize(negative_loglik, interval = c(0, 1))
c(numerical = likelihood_fit$minimum, analytic = 32/50,
  negative_loglik = likelihood_fit$objective)
#>      numerical      analytic negative_loglik
#>    0.6400059479    0.6400000000    2.1465474559
negative_loglik(.5)
#> [1] 4.13299674
```

`dbinom()` evaluates the binomial probability for the fixed count 32. With `log = TRUE`, it computes the log probability directly. The numerical and analytic estimates both round to .6400. The small numerical difference from exactly .64 reflects the search tolerance. At the fitted probability the negative log-likelihood is 2.1465, compared with 4.1330 at $p = .5$. The smaller value is the better fit according to this criterion. The parameter estimate .6400 is a completion probability; the criterion value 2.1465 is not.

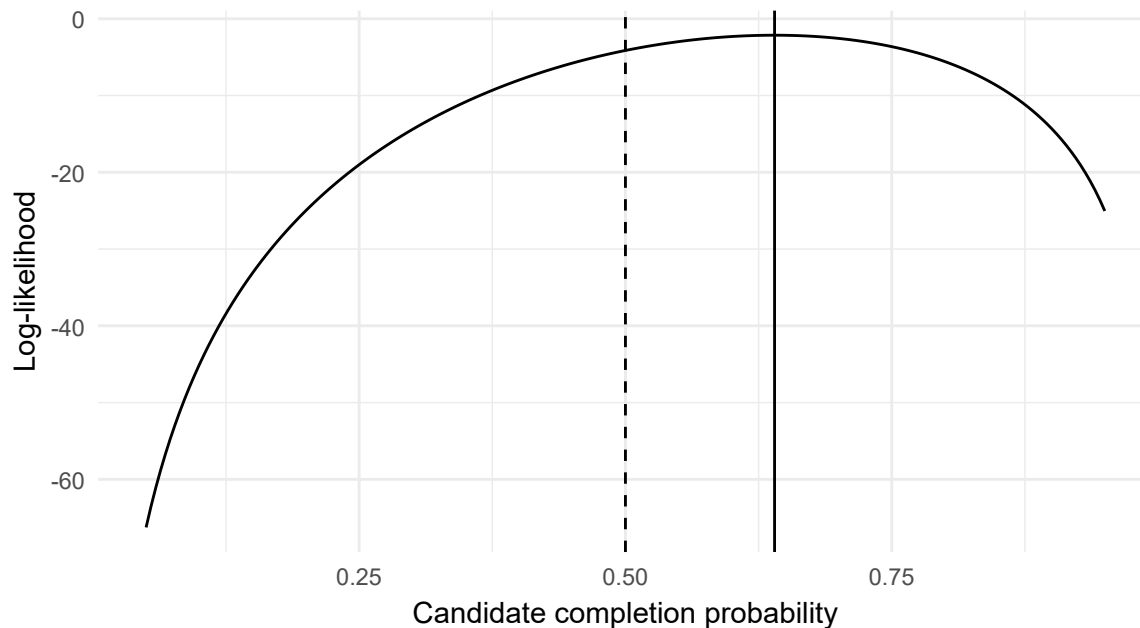


Figure 5: The binomial log-likelihood is largest at the observed proportion.

The horizontal axis contains candidate parameter values, and the vertical axis gives the log-likelihood of the same observed count at each candidate. Thus the curve displays a fitting criterion, not a sampling distribution or a distribution of probabilities over p . The peak identifies the fitted value. The score test and interval in Section 5 answer additional questions about a proposed null and the uncertainty of estimation under their stated calibration.

Finally, check the boundaries of the parameter space. With both completions and non-completions observed, $p = 0$ and $p = 1$ each give likelihood zero, so the maximum is interior. If all 50 trials instead had zero completions, the maximum would be at $p = 0$; if all completed, it would be at $p = 1$. An interior numerical search can approach such an endpoint without returning it exactly. The endpoint values must therefore be examined explicitly in those cases. Chapter 11 develops these boundary questions further when likelihood is used to fit logistic regression.

7 Examine procedures by repetition and plan precision

The preceding analyses started with observed data and produced estimates, intervals and tests. We can also begin with a declared generating model, create data from it, and examine what an analysis does over repeated studies. Chapter 5 used this approach to examine sampling

variation and interval coverage. Here the questions concern an omnibus ANOVA procedure and the sample size for a specified comparison.

7.1 Follow one generated study before repeating it

Imagine three independent groups whose outcomes follow Normal distributions, each with standard deviation $\sqrt{10/3} = 1.8257$ points. First set all three population means to 8, so that the equal-means null hypothesis is true. These values define a hypothetical world in which we know the truth. The constructed feedback scores used earlier help motivate the scale of the example; they do not establish that this generating model describes real pupils.

Begin with four observations per group and analyse one generated study.

```
set.seed(60630)
n_per_group <- 4
sim_group <- factor(rep(1:3, each = n_per_group))
sim_score <- rnorm(3*n_per_group,
  mean = rep(c(8, 8, 8), each = n_per_group), sd = sqrt(10/3))
anova(lm(sim_score ~ sim_group))
#> Analysis of Variance Table
#>
#> Response: sim_score
#>           Df      Sum Sq   Mean Sq F value    Pr(>F)
#> sim_group  2   7.597716  3.7988578  1.03652  0.39345
#> Residuals  9  32.985075  3.6650083
```

`set.seed()` makes this particular sequence of generated observations reproducible. `rep(1:3, each = n_per_group)` creates four labels for each of three groups, and `factor()` tells the model to treat those labels as categories. The second `rep()` supplies the population mean for each of the 12 observations. `rnorm()` draws independent outcomes with those means and the declared common SD.

The formula `sim_score ~ sim_group` fits a separate mean for each group; `anova()` compares that fit with a common mean. Its table has 2 group degrees of freedom and $12 - 3 = 9$ error degrees of freedom. For these generated data, $SS_{\text{between}} = 7.5977$ and $SS_{\text{within}} = 32.9851$, giving $F = 1.0365$ and $p = .3935$. Applying the rule $p < .05$ therefore does not reject the equal-means null in this study. Although the three generating means are identical, the three sample means vary, so the between-group sum of squares is positive. The estimated error variance, $MS_E = 3.6650$, also varies around the generating variance $10/3$.

One generated study cannot describe a procedure's long-run behaviour. The Computer Lab puts the same calculation in a small function: its arguments are the group size, the three

generating means and the common SD; its result is one omnibus p-value. Repeating that function gives a collection of p-values to compare with .05.

Table 9 reports 2,000 studies for each of four conditions. The null conditions use means (8, 8, 8); the alternative conditions use (6, 8, 10). Each condition retains independent Normal observations and the same common SD. For reproducibility, the lab restarts at seed 60630 and runs the conditions in this order: four per group under the null, four under the alternative, 12 under the null, then 12 under the alternative, without resetting the seed between conditions.

Table 9

Operating characteristics in the declared generating models, $B = 2000$ studies per condition.

Pupils per group	Null rejection rate	Power at means		
		MCSE	6/8/10	MCSE
4	.0560	0.0051	.6545	0.0106
12	.0500	0.0049	.9990	0.0007

The **null rejection rate** estimates the Type I error probability: how often this .05 omnibus procedure rejects when all three population means are equal. Both estimated rates are close to .05. The **power** column estimates the probability of rejecting that same null under the specified unequal means. It rises from .6545 with four pupils per group to .9990 with 12. This is power against the particular means 6, 8 and 10, with the stated spread and independence; a smaller separation would give different power.

Each rate is itself an estimate based on finitely many repetitions. If $\hat{q}$ is a simulated rejection proportion from B independent studies, its approximate Monte Carlo standard error is

$$MCSE(\hat{q}) = \sqrt{\frac{\hat{q}(1 - \hat{q})}{B}}.$$

For the estimated power .6545, the calculation is:

```
sqrt(.6545 * (1 - .6545) / 2000)
#> [1] 0.01063319684
```

The result, 0.0106, describes the numerical uncertainty from using 2,000 simulations. A fresh set of 2,000 simulated studies can give a somewhat different power estimate. This MCSE is distinct from a pupil outcome's SD or the SE of a group mean or contrast within one study. A rate very close to one, such as .9990, also deserves care: only two of the 2,000 studies failed to reject, so its tiny plug-in MCSE rests on very few non-rejections.

Increasing the number of pupils n changes the information within a study, its error degrees of freedom and its ability to detect the specified difference. Increasing the number of repetitions

B makes the simulated rate more precise; it leaves the study being simulated unchanged. The Computer Lab develops these changes in the repetition task and its reference extension. Simulation describes a procedure under the declared model; agreement with its expected behaviour does not establish that the model fits a particular real study.

7.2 Plan for a specified comparison and precision target

The omnibus simulation asks whether some population group means differ. A future study may instead be planned to estimate or detect one particular difference. Return to the two comparisons with Rereading from the one-way example: Answer feedback versus Rereading, and Explanatory feedback versus Rereading. Keep this declared **two-comparison Bonferroni family**, with family significance level .05, even when calculating power for just one of its comparisons.

Suppose the comparison of interest has a population difference of $\delta = 2$ points. Plan equal group sizes n , independent Normal outcomes and common SD $\sigma = \sqrt{10/3}$. The planning SE for a difference of two group means is $\sigma\sqrt{2/n}$. All three groups contribute to the pooled error variance, so its degrees of freedom are $3(n-1)$, even though the contrast compares only two of the means.

Bonferroni assigns two-sided significance level $.05/2 = .025$ to each comparison. Dividing that amount between its two tails puts .0125 in each tail. The rejection cutoffs are therefore $-c$ and c , where

$$c = t_{1-.05/(2 \times 2), 3(n-1)} = t_{.9875, 3(n-1)}.$$

Under the zero-difference null, the test statistic has the central t distribution already used for the comparison. Under a specified nonzero difference, its distribution moves towards that difference. The **noncentral t distribution** describes this statistic when its numerator has a nonzero mean and its denominator still estimates the common spread. Its noncentrality parameter expresses the population difference in planning SE units:

$$\lambda = \frac{\delta}{\sigma\sqrt{2/n}}.$$

For this alternative, power is the area beyond the same two rejection cutoffs: $P(T < -c) + P(T > c)$ for a noncentral t variable with $3(n-1)$ degrees of freedom and noncentrality λ . These are probabilities over future studies, rather than a p-value from observed data. Both tails belong to the two-sided procedure, even when the planning difference is positive.

```

contrast_power <- function(n, delta = 2, sigma = sqrt(10/3)) {
  df <- 3*(n-1)
  critical <- qt(1 - .05/(2*2), df)
  ncp <- delta / (sigma * sqrt(2/n))
  pt(-critical, df, ncp = ncp) +
    pt(critical, df, ncp = ncp, lower.tail = FALSE)
}
round(c(n16 = contrast_power(16), n17 = contrast_power(17)), 4)
#>      n16      n17
#> 0.7793 0.8073

```

The function accepts a proposed group size and returns power. Its first `pt()` call obtains the lower-tail probability; the second obtains the upper-tail probability. At $n = 17$, the planning SE is 0.6262, the error degrees of freedom are 48, the critical value is 2.3139, and $\lambda = 2/0.6262 \approx 3.1937$. Summing the two tail areas gives power .8073. Checking successive integer sizes gives 17 pupils per group, 51 in total, as the first size reaching .80 power; 16 gives .7793.

This is the power of **one specified comparison within the declared family**. It is neither the probability that both comparisons reject nor the power of the omnibus F test. For example, with 12 pupils per group the specified two-point comparison has power .6328, whereas the omnibus simulation in Table 9 gives estimated power .9990 against the complete pattern (6, 8, 10). The hypotheses and rejection rules differ, so these numbers answer different questions.

A precision target concerns the interval's width rather than its probability of excluding zero. Suppose the aim is a simultaneous interval whose planned half-width is no more than one point. The family and degrees of freedom stay the same, giving

$$\text{planned half-width} = t_{.9875, 3(n-1)} \sigma \sqrt{\frac{2}{n}}.$$

This calculation substitutes the planning SD σ for the error SD that the future study will estimate. Evaluating the two sizes either side of the threshold gives:

```

n_width <- c(34, 35)
round(data.frame(
  n = n_width,
  planned_half_width = qt(1 - .05/(2*2), 3*(n_width-1)) *
    sqrt(2*(10/3)/n_width)
), 4)
#>      n planned_half_width
#> 1 34              1.0078
#> 2 35              0.9929

```

The first group size meeting this planning target is 35, or 105 pupils in total: the planned half-width is 0.9929 points, compared with 1.0078 at 34. The realised interval uses the observed $\sqrt{MS_E}$ and therefore has a random width. Choosing 35 does not guarantee that its half-width will be at most one in every study. Figure 6 shows why a power target and an interval-width target can select quite different sample sizes.

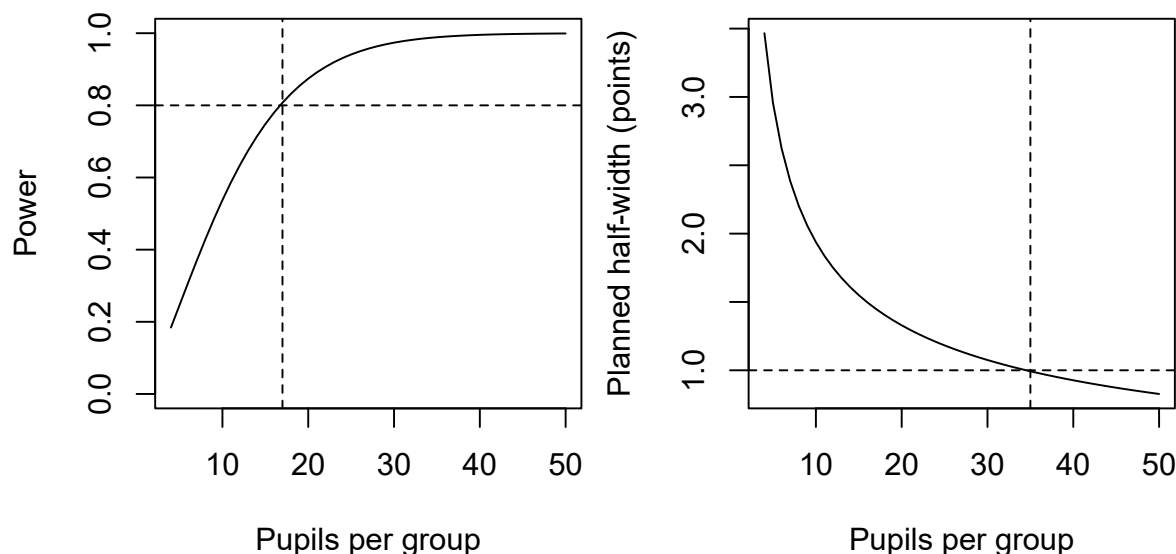


Figure 6: Power and planned interval width impose different sample-size targets.

Finally, distinguish the desired number of available outcomes from the number recruited. If outcomes are expected to be available for 90% of recruits, the 17-per-group power target gives:

```
ceiling(17 / .9)
#> [1] 19
```

Recruiting 19 per group gives 57 recruits and an expected 17.1 available outcomes per group under that retention assumption. It is an average allowance, not a guarantee of 17 outcomes in every group or a correction for bias from systematically missing outcomes.

The Computer Lab reference extension supplies the full sample-size grid and changes the target difference, comparison family and retention rate. Its two-group `power.t.test()` example declares a separate two-group study: that procedure has different error degrees of freedom and cannot silently replace this three-group calculation. Choose a planning difference from the substantive question and relevant prior evidence. As Chapter 3's significance-filter example showed, selecting statistically significant results can favour exaggerated estimates; an observed difference from such a selection is not automatically a suitable planning alternative.

8 Reading Bridge

The chapter compared group means, separated omnibus questions from calibrated contrasts, and read a comparison that changes across another factor. A separate baseline example showed what an adjusted comparison holds fixed. Counts required a different response model; likelihood and an objective function made the fitting criterion explicit. Repetition examined procedures under a declared generating model, while power and precision described different planning aims.

When reading a study, identify the question, analysed records, model, estimate and uncertainty procedure before evaluating their adequacy. For a group comparison, locate its subtraction direction and any covariates held fixed. For a test or interval, identify the null, reference distribution and comparison family. For a simulation or planning result, locate the generating assumptions and the particular procedure being evaluated. These details connect the reported numbers to the claims they can support.

Chapter 7 develops how a continuous-predictor line is fitted and read. Chapter 8 extends adjustment and distinguishes its conditional, standardized and predictive questions. Chapter 11 returns to likelihood for binary regression. Appendix B describes these connections more specifically.

Key terms and notation

ANOVA, contrasts, interactions and comparison families retain the meanings introduced in Chapter 4. The following concepts were introduced or developed through their R applications here.

Table 10

Selected concepts introduced or developed here.

Term	Definition
ANCOVA (analysis of covariance)	A linear model comparing groups while including quantitative covariates; its adjusted comparison is defined at stated covariate values or over a stated covariate distribution.
Model formula (R formula)	Notation such as <code>score ~ group</code> naming an outcome and the variables that explain it for a fitting function. It does not state whether observations are independent or paired; that comes from the design.
Objective function	A numerical criterion evaluated at candidate parameter values for fixed data; fitting seeks a stated minimum or maximum.
Likelihood and log-likelihood	The probability or density of the observed data under a declared model, viewed as a function of its parameters; the log-likelihood is its logarithm.
Maximum likelihood estimation (MLE)	Choosing parameter values that maximize the likelihood, equivalently the log-likelihood, over the stated parameter space.

The notation follows the local question. $C = \sum_j c_j \mu_j$ is a contrast, whereas $L(\theta)$ is a likelihood function of a parameter. Similarly, Type II error probability β differs from a model coefficient such as β_x . Observed Pearson X^2 differs from the reference distribution χ^2 . In simulations, n counts observations per stated study or group and B counts repeated studies. An SE describes an estimator's sampling variation; an MCSE describes finite-simulation uncertainty in a computed summary.

Further reading

For the linear-model basis of ANOVA, contrasts and covariance adjustment, see Kutner et al. (2005). Tukey (1953) develops simultaneous comparison ideas, and Wilson (1927) provides the score-interval construction for a proportion. Cohen (1988) discusses power and the dependence of a planning result on its chosen effect and procedure, and Çetinkaya-Rundel & Hardin (2024) develops the simulation-based treatment of operating characteristics used in Section 7. R's documentation gives the arguments and returned components for `aov`, `lm`, `TukeyHSD`, `oneway.test`, `prop.test`, `chisq.test` and `optimize` (R Core Team, 2025). The plotting

chapters of Wickham et al. (2023) develop the data, mapping and layer vocabulary used in the small `ggplot2` examples here.

Exercises

Use the chapter's printed data. Where a prompt supplies rounded summaries, answers within 0.02 in the stated units are sufficient unless that prompt requests a different precision. Keep each interpretation beside its target, denominator and working conditions.

Exercise 1: Read and reconstruct the one-way fit

Use the twelve Massed scores. Find the three means and grand mean. Reconstruct the between-group and within-group sums of squares, their degrees of freedom and the F statistic. State the equal-means null, explain what its p-value addresses under the working model, and calculate eta-squared.

Exercise 2: Keep the comparison and its family together

For the same scores, compare Answer feedback and Explanatory feedback separately with Rereading. Write both coefficient vectors, estimates and SEs. Construct the 95% Bonferroni intervals for these two comparisons and compare them with the corresponding all-pairs Tukey intervals. Explain why their limits differ.

Exercise 3: Read interaction as a change in comparison

Use all 24 feedback/practice scores. Produce the six cell means and the interaction plot. Calculate each feedback group's Spaced-minus-Massed change. Fit the full two-way model and report the interaction F test. For Answer's change minus Rereading's change, calculate the estimate, SE and 95% Scheffe interval over the two-dimensional interaction family. Explain how that interval can include zero when the omnibus interaction is significant.

Exercise 4: State what adjustment holds fixed

Use the separately printed twelve-pupil pretest/posttest data. Compare the raw group means with the fitted means at pretest 8 under the common-slope model. Reproduce the Answer-minus-Rereading estimate and its pointwise interval. Compare the baseline-only and full models on identical records. Explain the conditioning target, the common-slope assumption and why these authored results are not empirical evidence about feedback.

Exercise 5: Preserve the event, denominator and expected counts

In the invented table, Online has 32 completions and 18 non-completions; In person has 18 completions and 32 non-completions. For Online, test $p=.5$ with the uncorrected score procedure and report its 95% score interval. For the complete table, compute all expected counts and Pearson's X-squared statistic. Explain why the two tests answer different questions and what design information an effect claim would additionally need.

Exercise 6: Distinguish study size, repetitions and planning targets

Use the chapter's ANOVA simulation table. Explain the changes from $n=4$ to $n=12$ under the null and alternative models. Calculate MCSE for the .6545 power estimate with $B=2000$. Explain what increasing B changes. Then compare the $n=17$ power target with the $n=35$ planned-width target for the two-comparison family; say what each target controls and why a realised interval can have a different width from its planning value.

Exercise 7: Distinguish an estimate from its fitting criterion

Keep 32 successes in 50 trials fixed. Write the binomial likelihood as a function of p , find its analytic maximizer, and reproduce it using the negative-log-likelihood and `optimize()`. State the returned probability and objective value, compare the criterion at $p=.5$, and explain why a likelihood curve is not a probability distribution over p .

Exercise 8: Write a qualified result paragraph

Choose the one-way comparisons, selected interaction contrast or adjusted group comparison from this chapter. Write a short paragraph giving the question, data and denominator, estimate, interval and its family or pointwise status, working conditions and scope. Identify one tempting claim that the numerical result does not establish, and explain the missing support.

Appendix A: Alternative calculations on the same data

These short calculations expose relationships behind the main procedures. They use the objects created in the chapter. The Computer Lab references provide the corresponding variations and complete starting setups.

Direct sums of squares and omega-squared

The fitted object's residual sum of squares can be checked directly from the scores and group means. The following calculation reconstructs all three sums of squares and the observed eta-squared summary.

```
grand <- mean(massed$score)
ss_between <- sum(4 * (means-grand)^2)
ss_within <- sum(resid(one_fit)^2)
ss_total <- sum((massed$score-grand)^2)
c(SS_B = ss_between, SS_W = ss_within,
  SS_T = ss_total, eta2 = ss_between/ss_total)
#>          SS_B          SS_W          SS_T          eta2
#> 32.0000000000 30.0000000000 62.0000000000 0.5161290323
```

The values 32, 30 and 62 reproduce the main table, with $32 + 30 = 62$ and $\eta^2 = .5161$. For this one-way model, a conventional bias-adjusted variance-share estimator is

$$\hat{\omega}^2 = \frac{SS_B - (J - 1)MS_W}{SS_T + MS_W} = \frac{32 - 2(30/9)}{62 + 30/9} = .3878.$$

```
(ss_between - 2*mse) / (ss_total + mse)
#> [1] 0.387755102
```

Subtracting the error contribution from the numerator makes the estimate smaller here. In a sample with little between-group variation, that numerator can be negative. A negative estimate is a consequence of the bias adjustment, not a negative population variance share. Report the convention used if a display truncates it to zero. As with eta-squared, omega-squared does not replace the outcome-scale differences, their intervals or the design conditions for interpreting them.

Match the two-group t and F calculations

Restrict the Massed records to Rereading and Answer feedback. A pooled equal-variance t test and a two-group ANOVA then use the same eight pupils, two fitted means and six residual degrees of freedom.

```
two_groups <- droplevels(subset(massed,
  feedback %in% c("Rereading", "Answer feedback")))
pooled <- t.test(score ~ feedback, data = two_groups,
  var.equal = TRUE)
two_group_fit <- aov(score ~ feedback, data = two_groups)
c(t = unname(pooled$statistic),
  t_squared = unname(pooled$statistic)^2,
  p = pooled$p.value)
#>           t      t_squared          p
#> -1.5491933385  2.4000000000  0.1723082967
anova(two_group_fit)
#> Analysis of Variance Table
#>
#> Response: score
#>           Df Sum Sq   Mean Sq F value    Pr(>F)
#> feedback    1      8 8.0000000      2.4 0.17231
#> Residuals    6     20 3.3333333
```

The t statistic is -1.5492 because this formula call subtracts Answer's mean from Rereading's. Its square is $F = 2.4000$, and both tests give $p = .1723$. Squaring loses the sign but preserves the evidence against equal means for this matched procedure. Welch is a different calibration and does not generally share this identity; the equal group sizes and equal observed variances here happen to give the same numerical result. Using different records or the three-group residual denominator changes the calculation being compared.

Change the interval calibration for the selected DD

The selected Answer-minus-Rereading practice DD stays 4 with SE 1.8257. Only the multiplier changes between its pointwise t interval and the simultaneous Scheffe interval for the two-dimensional interaction family.

```
dd + c(-1, 1) * qt(.975, 18) * se_dd
#> [1] 0.16425869 7.83574131
dd + c(-1, 1) * scheffe * se_dd
#> [1] -0.8679613431 8.8679613431
```

The pointwise interval [0.1643, 7.8357] excludes zero; the simultaneous interval [−0.8680, 8.8680] includes it. Selecting the narrower interval after seeing which conclusion it yields does not preserve the broader family’s guarantee. Choose and describe the inferential question and calibration.

For comparison, Answer’s simple practice effect uses only two means:

```
simple_answer <- cell_mean["Answer feedback", "Spaced"] -
  cell_mean["Answer feedback", "Massed"]
se_simple <- sqrt(mse_two * (1/4 + 1/4))
c(estimate = simple_answer, SE = se_simple)
#>      estimate      SE
#> 4.000000000 1.290994449
simple_answer + c(-1, 1) * qt(.975, 18) * se_simple
#> [1] 1.287721309 6.712278691
```

Its pointwise interval is [1.2877, 6.7123]. The smaller SE reflects a different contrast. It does not acquire a simultaneous guarantee over simple effects from the interaction-only Scheffe calculation.

Compute the Wilson endpoints directly

For one proportion, write $z = z_{.975}$, $n = 50$ and $\hat{p} = 32/50$. Inverting the score inequality gives a center and half-width

$$\text{center} = \frac{\hat{p} + z^2/(2n)}{1 + z^2/n},$$

$$\text{half-width} = \frac{z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}.$$

```
phat <- 32/50
z <- qnorm(.975)
center <- (phat + z^2/(2*50)) / (1 + z^2/50)
half <- z * sqrt(phat*(1-phat)/50 + z^2/(4*50^2)) /
  (1 + z^2/50)
center + c(-1, 1)*half
#> [1] 0.5014101688 0.7586125035
```

The endpoints [.5014, .7586] reproduce the uncorrected one-proportion `prop.test()` interval. The altered center and half-width are consequences of score-test inversion. The denominator and Bernoulli working model remain those of the original 50 Online records.

Appendix B: What later chapters develop

Chapter 7 extends squared loss from a fitted center to a line. Its appendix recalls the mean-loss identity, derives the line’s normal equations and relates fitted values and residuals through projection. It also develops uncertainty for the slope and fitted mean. The scalar numerical search here supplies a first fitting example; those later derivations explain the linear model’s structure.

Chapter 8 extends adjustment to several controls. It distinguishes a coefficient conditional on specified predictors from a fitted mean at a profile, a prediction for a new case, and an average of fitted values over a named population. Coefficient covariances help calculate uncertainty for these quantities. The chapter also develops common support and the difference between observational and randomized-design interpretations. Chapter 9’s separate Baumann ANCOVA reference retains the original study’s outcome definitions, observed denominators and missing-data qualifications; it is distinct from the twelve authored pupils analysed here.

Chapter 11 develops likelihood for binary regression. Its reference material works through exact boundary estimates and their logit limits, and a Normal likelihood with more than one parameter. Numerical fitting then requires attention to parameter constraints, starting values and convergence checks. The scalar 32-of-50 calculation here introduces the objective and its maximizer while keeping those more extensive issues with their later applications.

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